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Identification of Gene-Therapy Responsive Blood Biomarkers in *mdx* Mouse Model

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Keywords: antibodies | Becker muscular dystrophy | biomarkers | Duchenne muscular dystrophy | plasma proteomics | rare disorders | suspension bead array

ABSTRACT

Introduction: Identifying serum biomarkers that reflect the restoration of dystrophin in skeletal muscle is important for evaluating the effect of dystrophin-restoring therapies in preclinical and clinical trials. Many potential blood biomarkers have been identified in Duchenne muscular dystrophy (DMD) patients, which change with disease progression or respond to pharmacological treatment. In this study, it was suggested that a panel of such blood biomarker candidates could be used to monitor dystrophin rescue in *mdx* mice treated with microdystrophin based therapies.

Methods: Plasma samples from *mdx* mice treated with the microdystrophin therapy SGT-001 were analysed with an antibody suspension bead array consisting of 87 antibodies. The array targets 83 unique proteins previously identified as biomarker candidates for DMD. Each sample was assayed at two different plasma dilutions to cover a broader concentration range. Protein concentrations estimated as Median fluorescent intensities (MFI) were correlated to dystrophin expression in muscle tissue, as measured by immunohistochemistry and Western blot. Thirteen of the targets were selected and analysed in a DMD and Becker muscular dystrophy (BMD) longitudinal natural history cohort using a suspension bead array.

Results: Ten proteins were found to be significantly elevated in untreated *mdx* mice compared with C57 wild-type mice and to correlate with dystrophin expression (Spearman's correlation, FDR < 0.05) upon gene transfer in *mdx* mice. Translatability of these biomarkers from animal models to patients was evaluated by exploring abundance trajectories over time in both DMD and BMD patients and association with dystrophin expression in BMD patients. Consistent with the observations in mouse, six of these biomarker candidates were more abundant in DMD patients compared with controls, and six were also differentially abundant between BMD and DMD patients. Among them, serum titin was shown to be associated with dystrophin expression in BMD patients, having a steeper decline over time in patients with low dystrophin expression in tibialis anterior compared with patients with high expression. Myosine light chain 3 had a steeper decline with time in DMD patients compared with BMD patients.

Conclusions: The 10 biomarker candidates identified in mouse plasma are related to muscle contraction, glycolysis, microtubule formation and protein degradation. Human titin and myosine light chain 3 were the most interesting candidates as explorative biomarkers to monitor microdystrophin expression in gene therapies. If confirmed, these biomarkers could be used to detect

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restoration of dystrophin expression per se, monitor changes in dystrophin expression over time and potentially confirm disease phenotype changes from severe to mild disease forms.

1 | Introduction

Duchenne muscular dystrophy (DMD) is a rare, currently incurable, fatal neuromuscular disorder with paediatric onset [1]. Different pathogenic variants (most commonly frame-shift mutations) within the *DMD* gene result in lack of expression of the dystrophin protein. Consequently, these patients suffer progressive and fatal muscle wasting, cardiorespiratory complications and cognitive comorbidities in a proportion of patients. Several clinical trials (ClinicalTrials.gov: NCT03333590, NCT04240314, NCT05429372, NCT04281485, NCT05096221 and NCT03368742) currently explore the efficacy of therapies aiming to deliver a shortened dystrophin gene, and one such microdystrophin therapy recently gained accelerated Food and Drug Administration (FDA) approval [2]. Replacement therapies aim to slow down the disease progression by providing patients with a short but potentially functional version of the *DMD* coding sequence. Shorter but functional dystrophin proteins are expressed in patients with Becker muscular dystrophy (BMD) as a consequence of in-frame deletions of the *DMD* gene. BMD patients experience a milder disease form with an increased life expectancy compared with DMD patients [3].

The skeletal muscle isoform of human dystrophin is a 427-kDa rod-shaped protein connecting the sarcolemma with the cytoskeleton, where it is proposed to act as a flexible stabilizer of the sarcolemma, protecting muscle fibres from the sheering stresses of contraction and relaxation [3]. The absence of dystrophin causes micro-tearing to the membrane and changes in calcium homeostasis, which in turn leads to altered cellular energy metabolism, calcium-dependent protease activation and apoptosis [3]. Muscle tissue regeneration is impaired over time, causing permanent loss of muscle mass, adipocyte infiltration into muscle tissue and fibrosis formation [4].

In the advent of dystrophin-restoring therapeutic intervention, investigators face several challenges. One such challenge is the large size of the *DMD* gene. Although delivery of dual hybrid vectors to express the full-length dystrophin protein in cell and animal models has been successful [5], ongoing clinical trials focus so far on the delivery of small dystrophin-like genes (microdystrophin genes), encoding exclusively for a selection of domains anticipated to obtain a functional protein [6]. Another challenge is that primary endpoints for clinical trials rely on clinical scale measurements, such as the North Star Ambulatory Assessment (NSAA) and 6-min walk test (6MWT), that require a thorough evaluation of clinically relevant threshold values to aid interpretation of clinical trial results [7]. DMD therapies, such as Elevidys [8] and Golodirsen [9], have received accelerated approval by the FDA based on partial restoration of dystrophin expression in muscle biopsies 'as it is reasonably likely to predict clinical outcome'. Low or heterogenous dystrophin expression [10] and the existence of revertant fibres [11] in some patients complicate assessment of dystrophin as treatment outcome. Standardized antibody-based [12] and mass spectrometry-based strategies [13] are therefore being developed in an attempt to improve accuracy when quantifying the dystrophin protein in muscle tissues. However, muscle biopsies provide only local data within a fraction of the muscle, and concerns have been raised by patients regarding the scarring and the infection risk from repeated biopsies [14].

With these challenges in mind, researchers have suggested [1, 15] that reliable biomarkers could be used as complementing exploratory endpoints to evaluate systemic pharmaco-gene therapy efficacy for microdystrophin, preferentially from low-invasive biofluids like blood or urine. Blood is especially attractive as proteins produced within muscle tissue end up in the blood stream and provide a more representative view of the entire muscle organ, in comparison to a tissue biopsy that is retrieved locally. A growing number of potential blood biomarkers for DMD have been identified over the years [1]. However, only a limit number of these have been explored in the context of evaluating pharmacodynamic effect [15–18].

In this study, it was postulated that one or several blood protein biomarkers can be used to monitor dystrophin and microdystrophin expression less invasively than using muscle biopsies. The goal was to propose blood biomarker candidates of gene-therapy efficacy as a first step in developing explorative biomarkers. To identify biomarker candidates, plasma samples were evaluated in *mdx* mice treated with increasing doses of a microdystrophin therapeutic (SGT-001), an adeno-associated virus (AAV)-based gene therapy. An antibody suspension bead array (SBA) consisting of antibodies towards 83 candidate proteins was used to analyse the blood samples. Top candidates were further evaluated in human samples obtained in longitudinal natural history cohorts in dystrophic patients. Trajectory changes of protein levels over time in DMD and BMD patients were investigated to assess the relationship between dystrophin levels in muscle and abundance of blood biomarkers.

2 | Materials and Methods

2.1 | Ethics

The animal study was performed at an Association for the Assessment and Accreditation of Laboratory Animal Care (AAALAC) accredited laboratory with an Animal Welfare Assurance, United States Department of Agriculture registration and compliance and regulation by an Institutional Animal Care and Use Committee (IACUC). This study was performed according to Declaration of Helsinki and approved by the Leiden University Medical Center (LUMC) *Commissie Medische Ethiek* (protocol NL50171.058.14), and the study was approved by the *Regionala etikprövningsnämnden Stockholm*, Sweden (ref. 2018/1859-31/1).

2.2 | Mouse Study Design and Sample Collection

To study the effect of microdystrophin on plasma biomarkers, plasma and paired microdystrophin expression was selected

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across six studies to obtain a wide range of doses and expression. Five-week-old *mdx* C57BL/10ScSn-Dmdmdx/J male mice were obtained from the Jackson Laboratory (Bar Harbor, ME, USA) in each of those six studies. The mice were fed with standard mouse chow and water *ad libitum* and kept under a 12-h:12-h light:dark cycle. The *mdx* mice in each of these studies were administered SGT-001, an AAV-9 vector containing a muscle-specific promoter and microdystrophin, as a single intravenous bolus dose to 6-week-old mice (Day 1) after 1 week of acclimatization at doses 1.6E12, 7.8E12, 1.6E13, 2.4E13, 5E13, 1E14 and 2E14 vg/kg (Table 1). Blood was collected prior to necropsy on Day 29 for treated *mdx* mice and was banked as plasma after centrifuging at 2200 × *g* for 15 min at room temperature within 45 min of collection and stored at −80°C. Because this was a retrospective study, plasma from wild-type (C57BL/10ScSn/J) and *mdx* controls (12 weeks of age) were supplemented to provide disease controls. Following necropsy and cryopreservation, quadricep muscles were analysed using immunofluorescence to quantify the membrane-localized microdystrophin protein. Isopentane frozen muscles were sectioned (8 μm) and stained for microdystrophin using MANEX44A (DSHB). The percentages of positive fibres were reported based on visual inspection by a board-certified pathologist. Further details on the method have been previously published [19].

2.3 | Human Natural History Study Design and Sample Collection

Patients of age ≥ 18 years diagnosed with BMD ($n = 34$) were recruited via the Dutch Dystrophinopathy Database for a 4-year prospective natural history studied conducted at the LUMC [20]. Previously collected samples were also included if available from the local biobank. This led to the inclusion of one extra serum sample for 17 patients. Annually, BMD patients performed functional tests, for example, NSAA, 10-m walk/run test velocity (10MWT), and the 6MWT. Yearly serum samples were collected and prepared as described by van de Velde et al. [20] and stored at −80°C prior use. Muscle biopsies of the tibialis anterior (TA) muscle were collected and dystrophin quantified by capillary Western immunoassay in duplicates using two independent antibodies according to a previously published protocol [21]. Dystrophin expression was estimated as percentage relative α-actinin. Reliable and precise quantification of dystrophin was performed using two independent antibodies previously reported [21]. DMD patients ($n = 19$) were recruited, and 49 samples were collected longitudinally as part of research protocol P05.004 approved by LUMC Institutional Review Board.

2.4 | Target and Antibody Selection

A list of 95 biomarker candidates were constructed comprising serum proteins differentially abundant between wild-type (wt) and *mdx* mice, and proteins with a restoration towards wt levels in a dose responsive manner [22]. For each target, antibodies were selected from the Human Protein Atlas [23] repository based on three criteria: (i) Sequence similarity of the recombinant Protein Epitope Signature Tag (PrEST) [23] used as immunogen had to be above 70% to mouse protein homologue. (ii) Antibodies should be successfully validated on antigen array and Western Blot [24]. (iii) Antibody concentrations had to be above 0.05 mg/mL. Following filtering, 87 antibodies were selected towards 83 targets. Monoclonal antibodies MANDYS106 (targeting dystrophin) was purchased from Sigma Aldrich, and AMAb90516 (vimentin) was purchased from Atlas Antibodies. For full selection of targets and antibodies, see Supporting information S1.

2.5 | Construction of Antibody SBAs

Antibodies were coupled to magnetic carboxylated beads with unique colour identifier (ID) to create the antibody SBAs and subsequently used in the immunoassays to detect the targeted proteins in serum or plasma. The SBAs were constructed as previously described [25], but with some modifications (see Supporting information S2 for antibodies and blocking buffer for each SBA). Two negative controls were included in each SBA; one bead ID coupled to 1.09-μg bovine serum albumin fraction V (Merck KGaA) or polyclonal rabbit IgG isotope control (Bethyl) and one uncoupled (empty) bead ID. Coupled beads

TABLE 1 | Study design and data from dystrophin expression measurements in quadricep muscles.

Mouse strain	Treatment	Dose (vg/kg)	Dystrophin level (mean ± SD, %)	*N* individuals
mdx	Saline	NA	0 ± 0	5
mdx	SGT-001	1.60E+12	9 ± 11	4
mdx	SGT-001	7.80E+12	16 ± 8	4
mdx	SGT-001	1.60E+13	26 ± 24	4
mdx	SGT-001	2.40E+13	3 ± 0	2
mdx	SGT-001	5.00E+13	31 ± 21	9 (1)[a]
mdx	SGT-001	1.00E+14	30 ± 14	4
mdx	SGT-001	2.00E+14	78 ± 15	6
wt	Saline	NA	100 ± 0	5 (2)[a]

[a]Number of samples in parenthesis were removed in downstream analysis due to technical issues during sample preparation.

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were blocked overnight at 8°C in either 5% (w/v) bovine serum albumin fraction V in PBST (137 mM NaCl, 2.7 mM KCl, 10 mM Na_2HPO_4, 1.8 mM KH_2PO_4, 0.05% Tween 20, pH 7.4) or PVXcas buffer containing 0.5% (w/v) polyvinylalcohol (Merck KGaA), 0.8% (w/v) polyvinylpyrrolidone (Merck KGaA) and 0.1% (w/v) and casein (Sigma-Aldrich) in PBS (137 mM NaCl, 2.7 mM KCl, 10 mM Na_2HPO_4, 1.8 mM KH_2PO_4, pH 7.4).

2.6 | Biotin Labelling and Multiplex Immunoassay of Plasma or Serum

Serum and plasma samples were biotin labelled, heat treated and incubated with antibody labelled beads according to a previously published protocol [25]. The beads were washed to remove unbound proteins, incubated with fluorophore (R-phycoerythrin conjugated streptavidin, Invitrogen) and subsequently analysed using the Luminex 100/200 System flow cytometer (Luminex Corp.) to retrieve mean florescence intensity (MFI) for each individual bead ID [25]. Pooled samples, created by mixing equal volumes of each serum or plasma sample included in the study were used for optimization experiments to establish biotin labelling conditions and sample dilutions [25]. Unprocessed $\log_2$(MFI) signals obtained from the analysis with the Luminex 100/200 were evaluated against total protein concentration for each analyte. Antibody-target interaction was further manually classified as being either (i) below the linear range, (ii) within the linear range or (iii) above the linear range for target concentrations within the pooled sample. The total plasma or serum protein concentration was measured for each sample as well as for the pooled sample using Pierce Bichiconic acid assay Protein Assay Kit (Thermo Scientific) according to the manufacturer's instructions.

Samples were randomized and diluted in PBS to two different end dilutions, biotin labelled and assayed together with SBAs as previously described [25], but with some modifications. Mouse plasma samples were diluted in an assay buffer containing 0.5 mg/mL rabbit IgG 1:1000 ProClin 300 (Sigma-Aldrich) PBST for overnight incubation with beads. Human serum samples and a human serum pool were labelled at between 1.5 and 8.6 mg/mL serum to evaluate effect of protein to biotin ratio on assay performance. As a third dilution control, the final assay was performed at two different end dilutions (0.172 mg/mL labelled serum and 0.258 mg/mL labelled serum) with a pooled sample diluted between 0.025 and 0.344 mg/mL labelled serum. All assays were analysed on a Luminex 100/200 instrument using xPONENT software (Luminex Corp.).

2.7 | Data Pre-Processing and Statistical Analysis

Obtained raw MFI were $\log_2$-transformed and technical outliers were removed prior to data analysis. Technical outliers were identified in three steps: Sample colour was assessed in order to exclude samples with high degree of haemolysis and contamination with haemoglobin; principal component analysis (PCA) was used on data obtained before and after repeating biotin labelling to detect changes in samples clustering due to the biotin labelling process; and MFI signals for fibrinogen subunit gamma (FGG) were manually evaluated to identify samples that appeared to have self-clotted during plasma preparation. After removing outliers, MFI signals were normalized, to minimize variations in sample dilution, using a modified version of probabilistic quotient normalization (PQN) [26] where sample-specific scaling factors were calculated from antibodies hypothesized to represent background noise. Antibodies were classified as background based on two main criteria: antibodies with signals that correlated to the negative control beads in both conditions with a Spearman's rank correlation coefficient above 0.8 (Figure S1) and those with signals below or within the linear detection range for pooled samples.

All statistical analysis was performed using R software 4.1.0 [27]. Median fluorescent intensity (MFI) signals obtained for each target on mouse plasma and human serum cohorts were first filtered to remove signals with high intra-well covariance (Figure S2), then $\log_2$-transformed and normalized through modified PQN [26]. The batch effect between multiple SBAs was reduced using the ComBat algorithm from R package *sva* [28]. All *p* values were corrected for multiple hypotheses testing using Benjamini–Hochberg procedure and presented as false discovery rates (FDR) [29].

For the mouse plasma cohort, a fold change (FC) filter was applied on normalized data after each statistical test to only include proteins with a mean $|\log_2 FC| > 0.38$, corresponding to approximately three times the technical variation. This was done in order to avoid over-interpreting proteins with near-background behaviour. The *p* values were corrected for multiple hypothesis tests before applying the fold change filter.

For the patient serum cohort, linear mixed effects models were used to investigate the relationship between tissue expression of dystrophin in BMD patients and biomarkers in serum. The model considered time (the number of years between the time of sample collection and the first visit), dystrophin expression in TA (%) and the interaction of the two as fixed effects and patient as random effect. Another linear mixed effects model was used to evaluate differences in biomarker trajectories between DMD and BMD patients over time by keeping diagnosis and time from first collected sample as fixed effects. Models were generated using R package *lme4* [30] Risk of model overfitting was evaluated by delta Akaike information criterion (ΔAICc) [31, 32] using the R package *bbmle* [33] and likelihood ratio test [34].

3 | Results

3.1 | Plasma Proteins Correlating With Dystrophin Rescue in *mdx* Mice

In this study, it was postulated that biomarkers could both be significantly different in plasma from *mdx* mice compared with plasma from wt mice with normal dystrophin expression, as well as correlate with degree of dystrophin rescue post treatment. To test this hypothesis, plasma samples from *mdx* mice treated with either vehicle or increasing doses of SGT-001 were analysed with an SBA consisting of 87 antibodies targeting 83 proteins. Two assay conditions (0.114 and 0.228 mg/mL total plasma proteins), based on results from preliminary optimizing experiments (Table S1, Figure S1), were included to ensure

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proteins would be detected within the dynamic range of each antibody.

Student's *t*-test identified differentially abundant proteins in *mdx* mice ($n = 5$) compared with wt mice ($n = 3$). Linear comparison (Figure 1A) and volcano plots (Figure 1B) show 10 proteins, proteasome subunit alpha type-2, triosephosphate isomerase (mPSMA2), adenylosuccinate synthetase isozyme 1 (mADSSL1), *N*-acylsphingosine amidohydrolase 1 (mASAH1), titin (mTTN), carbonic anhydrase 3 (mCA3), OTU domain-containing protein 5 (mOTUD5), mitochondrial LON protease 1 (mLONP1), nuclear migration protein nudC (mNUDC) and myosin light chain 3 (mMYL3), elevated in *mdx* mice with total protein concentration of 0.114 mg/mL (Table 2) and 5 proteins (mPSMA2, mTPI1, mADSSL1, mARHGEF10, and mASAH1) at 0.228 mg/mL plasma proteins concentration (Table S2). The reduced number of identified targets in the assay with the highest plasma concentration might indicate that levels of saturation are reached under these conditions. Notably, one protein, Rho guanine nucleotide exchange factor 10 (mARHGEF10), which had a low mean MFI and was not significant in the 0.114 mg/mL assay, was found significantly elevated in *mdx* mice in the 0.228 mg/mL assay (FDR < 0.05). Successful detection of this protein was improved in the high-concentration sample as the assay appeared to be close to its lower detection limit in the preliminary pooled sample assay, illustrating the advantage of performing an SBA at different sample dilutions. Heatmap analysis of protein *z*-scores revealed that about half of the analysed proteins share a pattern of being more abundant in *mdx* mice compared with wt mice (Figure 1C).

To explore in which context identified markers are useful, biomarker association with dystrophin restoration was investigated. Spearman's rank correlation analysis was performed to assess whether any of the detected proteins correlated with dystrophin recovery in muscle, by considering the wt mice to have 0% treatment induced dystrophin expression. Nineteen proteins were found to be significantly correlated (FDR < 0.05) with dystrophin expression in the 0.114 mg/mL assay (Table 2) while 10 proteins were identified in the high-concentration samples (Table S3); 9 were shared in both assay conditions. Ten of these proteins were both elevated before treatment and correlated with dystrophin rescue in either one or both assay conditions (Figure 2A). To further explore the functions of these proteins in humans (stated with the prefix h), a database search using the Human Protein Atlas (www.proteinatlas.org) [23] was performed. Tissue expression clustering and single-cell expression clustering showed that the corresponding 10 proteins in humans are involved in muscle contraction (hADSSL1, hASAH1, hCA3, hMYL3, hTTN), (hTPI1), and protein degradation (hPSMA2, hOTUD5, hLONP1). These proteins had either enhanced expression in skeletal and cardiac myocytes (hTTN, hCA3, hMYL3, hADSSL1, hASAH1, hTPI1), or localized to proteasome (hPSMA2) or mitochondria (hOTUD5, hLONP1) in most tissues.

Similarly, Spearman's rank correlation was used to identify proteins correlating with increasing therapeutic dose. Only *mdx* mice were considered for this analysis. At 0.114 mg/mL plasma, 26 proteins correlated with dose (Table 3), with 7 proteins also found to be upregulated in untreated *mdx* mice and 15 correlating with both dose and dystrophin expression (Figure 2B–C).

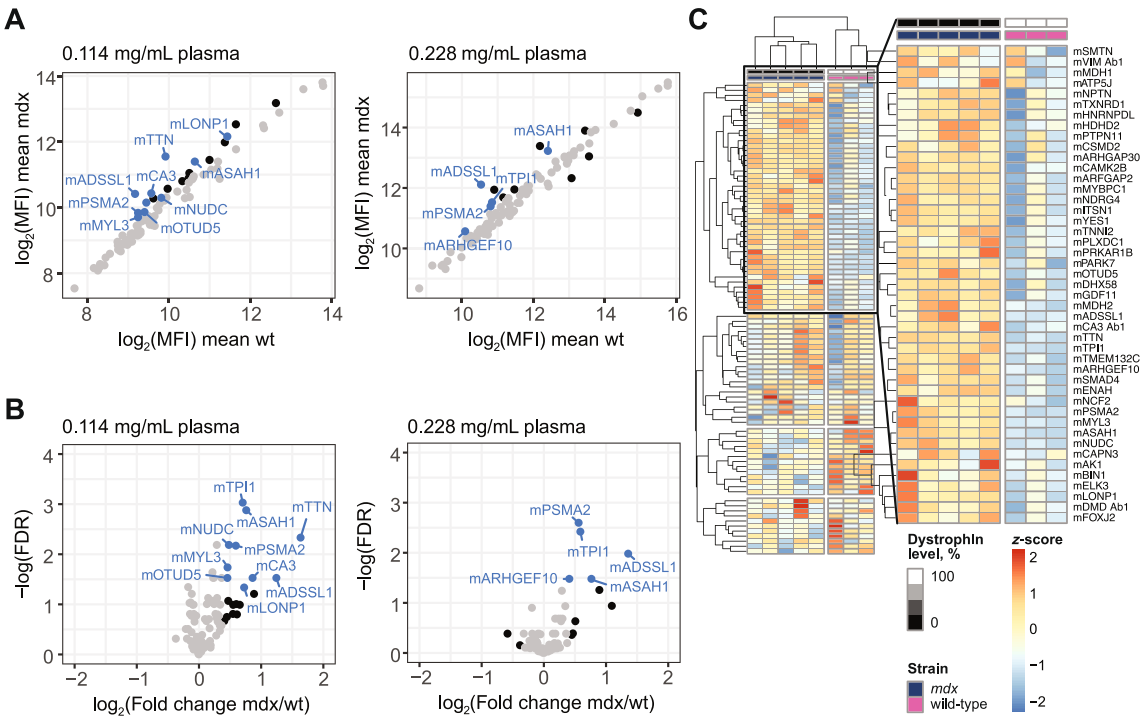


FIGURE 1 | Identification of proteins significantly differentially abundant between untreated *mdx* mice ($n = 5$) and wt mice ($n = 3$). (A) Linear comparison between group mean $\log_2$ (MFI) for *mdx* mice compared with wt mice. (B) Volcano plots for proteins differentially abundant in *mdx* mice compared with wt. Significantly (FDR < 0.05) differentially abundant proteins with a $|\log_2(FC)| > 0.38$ between group means are marked in blue. Nonsignificant proteins with a $|\log_2(FC)| > 0.38$ are marked in black and proteins with a $|\log_2(FC)| < 0.38$ marked in grey. (C) Heatmap for calculated *z*-scores for samples from *mdx* mice and wt mice show a cluster of proteins, which are elevated in *mdx* and capable of completely separating the two groups.

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TABLE 2 | Targets correlating with dystrophin expression (Spearman's correlation) and abundance difference between *mdx* and wt (Student's *t*-test) for the 0.114 mg/mL plasma assay.

		0.114 mg/mL plasma proteins						
					$\log_2$(FC)	Student's *t*-test		
Antibody	**Protein**	**Spearman's rho**	***p***	**FDR**	***mdx*/wt**	***mdx* vs. wt**	***p***	**FDR**
HPA030048	mTTN	−0.77	8.81E−09	8.02E−07*	1.63	13	1.53E−04	4.65E−03*
HPA008188	mPSMA2	−0.72	1.61E−07	7.31E−06*	0.59	8.3	4.45E−04	6.74E−03*
HPA017375	mOTUD5	−0.68	1.17E−06	3.46E−05*	0.46	5.4	3.27E−03	2.97E−02*
HPA052621	mADSSL1	−0.66	3.55E−06	5.38E−05*	1.24	5.5	4.42E−03	2.97E−02*
HPA005468	mASAH1	−0.63	1.29E−05	1.66E−04*	0.76	12.3	2.90E−05	1.32E−03*
HPA016649	mARFGAP2	−0.63	1.50E−05	1.66E−04*	0.60	5.3	2.73E−02	9.87E−02
HPA016564	mMYL3	−0.61	3.38E−05	2.55E−04*	0.46	7.1	1.41E−03	1.83E−02*
HPA018007	mITSN1	−0.60	3.64E−05	2.55E−04*	0.61	3.3	6.54E−02	1.59E−01
CA3.0121	mCA3	−0.59	5.47E−05	3.36E−04*	0.86	4.7	3.50E−03	2.97E−02*
HPA053568	mTPI1	−0.57	1.07E−04	5.42E−04*	0.70	14.7	1.02E−05	9.32E−04*
HPA027183	mNUDC	−0.57	1.35E−04	6.31E−04*	0.48	8.2	3.58E−04	6.51E−03*
HPA043871	mTXNRD1	−0.57	1.39E−04	6.31E−04*	0.41	2.3	1.11E−01	2.16E−01
HPA027614	mMYBPC1	−0.53	4.57E−04	1.73E−03*	0.55	3.5	6.12E−02	1.55E−01
HPA002034	mLONP1	−0.53	4.96E−04	1.78E−03*	0.73	4.5	8.09E−03	4.60E−02*
HPA012535	mPLXDC1	−0.52	5.09E−04	1.78E−03*	0.47	3.2	2.06E−02	8.52E−02
HPA015779	mNDRG4	−0.50	8.98E−04	2.69E−03*	0.45	2.9	8.44E−02	1.79E−01
HPA026307	mCAMK2B	−0.50	9.17E−04	2.69E−03*	0.66	4.4	3.36E−02	1.02E−01
HPA023885	mDMD	−0.43	5.13E−03	1.14E−02*	0.55	3.1	3.56E−02	1.02E−01
HPA055938	mTNNI2	−0.41	8.24E−03	1.67E−02*	0.89	4.3	1.22E−02	6.18E−02

Note: Only targets with a $|\log_2(FC)| > 0.38$ between each group mean were considered in order to avoid targets with high signal-to-noise ratio. The number of untreated *mdx* and wt samples were 5 and 3, respectively, after removing technical outliers.
*Significant at FDR < 0.05.

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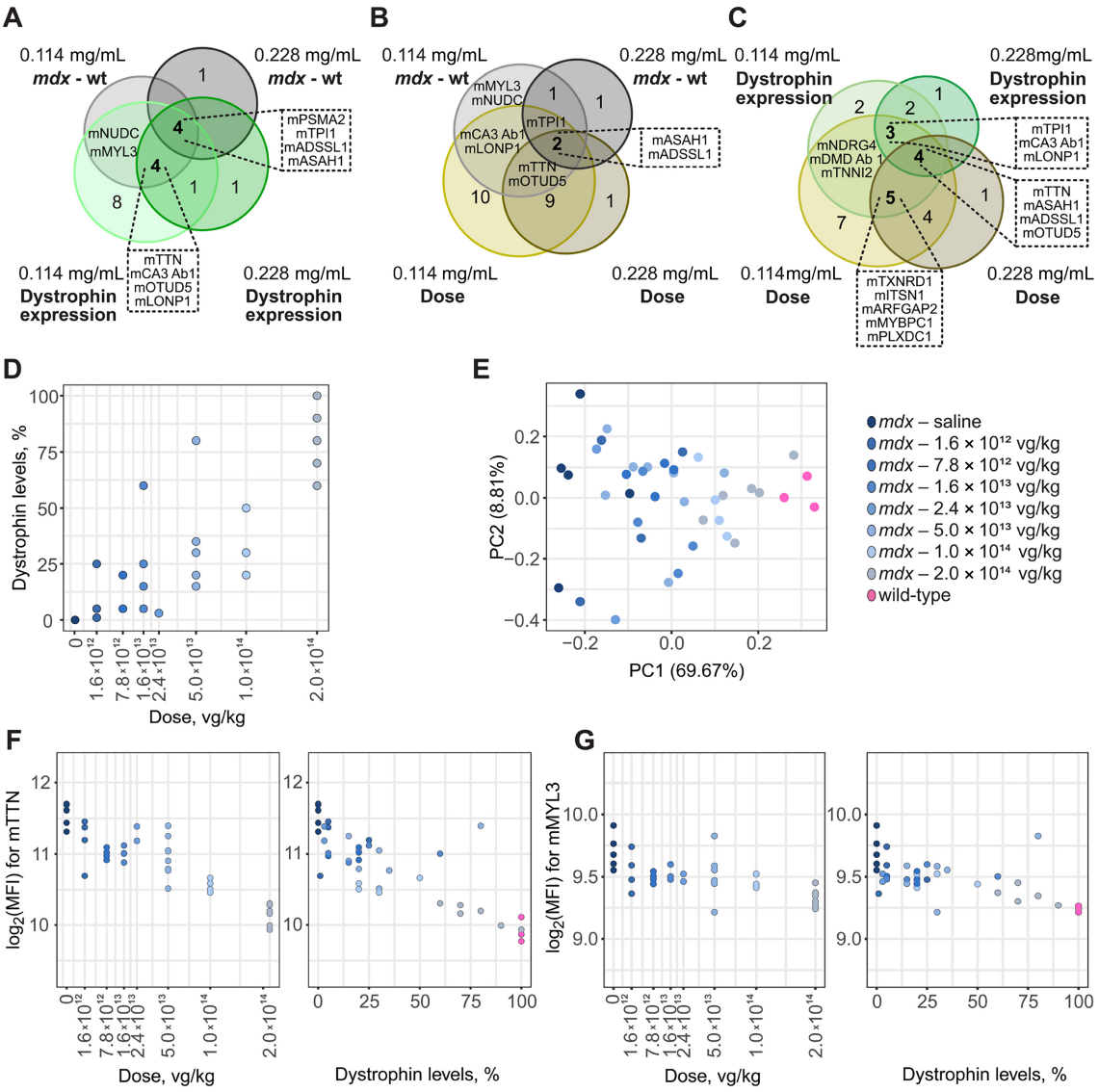


FIGURE 2 | Plasma proteins correlating with dystrophin expression, dose or both. (A–C) Venn diagrams describing overlap between proteins found to be significantly altered in (A) untreated *mdx* mice compared with wt (Student's *t*-test, FDR < 0.05) versus proteins found to correlate with increased dystrophin/microdystrophin expression (Spearman's correlation, FDR < 0.05), (B) untreated *mdx* mice compared with wt (Student's *t*-test, FDR < 0.05) vs dose (Spearman's correlation, FDR < 0.05) in treated *mdx* mice, or (C) proteins found to correlate with increased dystrophin/microdystrophin expression versus proteins found to correlate with dose in treated *mdx* mice. (D) Level of microdystrophin expression at each therapeutic dose interval. (E) Principal component analysis for the 10 proteins that both correlate with dystrophin expression and show a significantly different plasma abundance in untreated *mdx* compared with wt show a clear separation between different dose levels. (F,G) Correlation to dose and dystrophin expression for (F) TTN (antibody HPA030048) and (G) MYL3 (antibody HPA016564) in mouse plasma.

Because it was observed that the therapeutic dose is not always predictive of dystrophin rescue (Figure 2D), we considered proteins correlating with dystrophin rescue in muscle more informative. PCA performed on the 10 proteins (mPSMA2, mTPI1, mADSSL1, mASAH1, mTTN, mCA3, mOTUD5, mLONP1, mNUDC and mMYL3) showed that the first principal component (PC) can separate samples with different levels of dystrophin rescue (Figure 2E). Interestingly, the two proteins with the highest loadings in the first PC were mTTN and mMYL3, both important for muscle contraction. Figure 2F,G shows the correlation of mTTN and mMYL3 to dose and dystrophin rescue at 0.114mg/mL plasma. Out of the proteins considered, mTTN also had the highest fold change and the strongest correlation with dystrophin rescue. These 10 proteins, along with an antibody towards dystrophin (hDMD) and ATP-dependent RNA helicase DDX6 (hDDX6), were taken forward for confirmation in human serum. Previous studies [35] have hinted at the possibility of detecting dystrophin molecules in blood from healthy individuals which might reflect leakage from muscle. Thus, if it is possible to detect hDMD protein fragments in blood, it may pose a direct way of monitoring dystrophin rescue without the need of repeated muscle biopsies. For simplified reading purposes, we will from now on refer to the dystrophin protein detected in serum using an anti-DMD antibody as hDMD, whereas dystrophin expression in TA muscle tissue will not be abbreviated. The second protein, hDDX6, was included as a normalization control.

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TABLE 3 | Targets associated with dose of STG-001 as assessed through Spearman's correlation for the 0.114 mg/mL plasma assay.

		0.114 mg/mL plasma proteins			
					Log_2 (FC)
Antibody	**Protein**	**Spearman's rho**	***p***	**FDR**	**Dose 0 vs. 2E14 vg/kg**
HPA030048	mTTN	−0.81	8.46E−10	7.70E−08	1.41
HPA053568	mTPI1	−0.70	1.45E−06	3.31E−05	0.40
HPA043871	mTXNRD1	−0.65	1.44E−05	1.93E−04	0.66
CA3.0121	mCA3	−0.62	4.31E−05	3.02E−04	0.45
HPA005468	mASAH1	−0.62	4.93E−05	3.16E−04	0.52
HPA052621	mADSSL1	−0.61	5.22E−05	3.16E−04	1.04
HPA018007	mITSN1	−0.61	6.01E−05	3.42E−04	0.77
HPA016649	mARFGAP2	−0.61	9.22E−05	4.94E−04	0.52
HPA017375	mOTUD5	−0.59	1.31E−04	6.62E−04	0.46
HPA063250	mCCNH	−0.59	1.38E−04	6.62E−04	0.61
HPA027614	mMYBPC1	−0.56	3.04E−04	1.15E−03	0.75
HPA069674	mPTPN11	−0.55	3.75E−04	1.31E−03	0.68
HPA057380	mCOLEC11	−0.55	4.05E−04	1.37E−03	0.46
HPA015779	mNDRG4	−0.54	5.08E−04	1.59E−03	0.71
HPA036300	mARHGAP30	−0.52	8.78E−04	2.58E−03	0.52
HPA004190	mPARK7	−0.52	9.55E−04	2.72E−03	0.49
HPA002034	mLONP1	−0.50	1.62E−03	4.20E−03	0.54
HPA057153	mSNAI2	−0.49	1.99E−03	4.80E−03	0.44
HPA012535	mPLXDC1	−0.49	2.01E−03	4.80E−03	0.58
HPA053345	mCOL11A2	−0.49	2.06E−03	4.80E−03	0.68
HPA023885	mDMD	−0.49	2.26E−03	5.13E−03	0.39
HPA047749	mTRA2B	−0.44	5.96E−03	1.15E−02	0.4
HPA055938	mTNNI2	−0.44	6.27E−03	1.19E−02	0.39
HPA008622	mCSMD2	−0.42	1.05E−02	1.84E−02	0.54
HPA053886	mNRXN1	−0.39	1.78E−02	2.89E−02	0.44
HPA069365	mFLT4	−0.38	1.94E−02	3.10E−02	0.39

Note: Only targets with a $|\log_2(FC)| > 0.38$ between each group mean were considered in order to avoid targets with high signal-to-noise ratio.

3.2 | Ten Promising Biomarker Candidates Were Further Evaluated in a Natural History Cohort, Highlighting hMYL3, hDMD and hCA3 as Potential Markers of Dystrophin Rescue

To assess whether the relationships between dystrophin recovery in muscle and blood biomarkers observed in mouse translate to the human condition, we sought to investigate the identified proteins in a prospective natural history study involving BMD and DMD patients. We first investigated whether the proposed biomarker candidates had trajectories in relation to dystrophin expression in TA. Samples obtained from 34 male patients were analysed in this study. Baseline age was 39.9 years (SD 12.1). Twenty-six patients were carrying deletion (76.5%) while other patients were carrying either duplications or small variants. Samples of 19 DMD patients were also included. The baseline age was 7.4 years (SD 4.9). Eleven patients carried deletions (58%) while other patients carried duplications and small variants.

Patient samples were diluted to two different concentrations, 0.172 and 0.258 mg/mL serum. Twelve antibodies were used to target hPSMA2, hTPI1, hADSSL1, hASAH1, hTTN, hCA3, hOTUD5, hLONP1, hNUDC, hMYL3, hDMD and hDDX6 with an SBA. Antibody against amino acids 2843–2992 of the Dp427m isoform of dystrophin (hDMD) was used to detect hDMD in serum. This sequence is also found on the shorter hDMD protein isoforms Dp260, Dp140, Dp116 and Dp71, not enabling distinction between these isoforms in this assay. A linear

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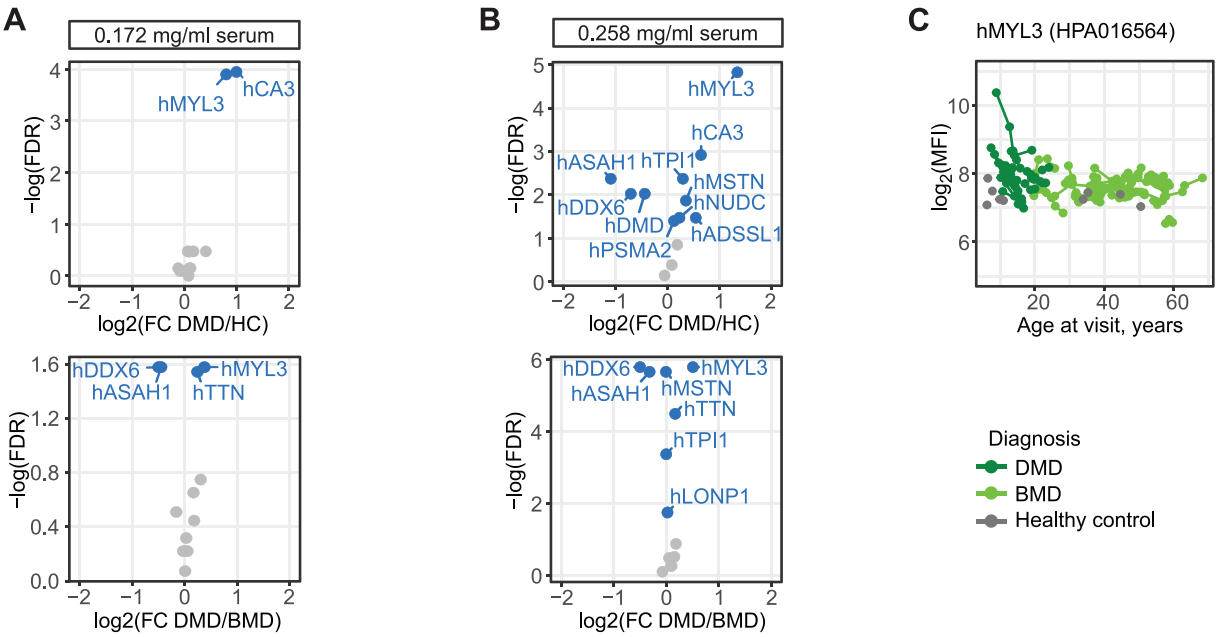


FIGURE 3 | Proteins significantly different between DMD, BMD and healthy controls. (A) Volcano plots for proteins with a significant difference between DMD and healthy controls or DMD and BMD as assessed through Wilcoxon ranked sum test. Only first sample collected from each patient is included in graph. Blue dots have FDR < 0.05 and $|\log_2(FC)| > 0.3$. (B) Human MYL3 was found to have significant change in age trajectory (FDR < 0.05) between DMD and BMD patients using linear mixed effect models. Samples from the same patient are connected with a line.

TABLE 4 | Test statistics from linear mixed effect model identify MYL3 with different age trajectory between DMD, BMD and healthy controls. MYL3 was analysed with hMYL3 (HPA016564) in 0.172 mg/mL serum.

Regression	Coefficient	False discovery	Rate
$\beta_{Intercept}$	8.11	$FDR_{Intercept}$	2.82E−73*
β_{time}	−0.12	FDR_{time}	3.68E−10*
β_{BMD}	−0.49	FDR_{BMD}	1.06E−03*
β_{HC}	−0.78	FDR_{HC}	2.20E−04*
$\beta_{time:BMD}$	0.11	$FDR_{time:BMD}$	5.62E−07*

Abbreviations: BMD, Becker muscular dystrophy; FDR: False discovery rate, estimated through the Benjamini–Hochberg method; HC, healthy controls; time, years from first visit in study.
*Significant at FDR < 0.05.

mixed effects model identified seven proteins, hOTUD5, hCA3, hTTN, hDDX6, hDMD, hADSSL1 and hASAH1 (Table S4) with abundance trajectories over time being associated with dystrophin expression in muscle (Figure S3). All seven proteins showed steeper decline in serum from patients with low TA dystrophin expression, relative α-actinin, compared with patients with higher expression. Likelihood ratio test confirmed that both time and dystrophin expression were important to include in the model and ΔAICc suggested that models were not overfitted. Evaluation of model parameters can be found in Supporting information S3 (model 1 to model 1.4).

Subsequently, we explored if these biomarkers can distinguish between patients lacking dystrophin expression in TA (DMD), patients with partial dystrophin expression (BMD) and healthy controls. The hMYL3, hCA3, hTPI1, hNUDC, hPSMA2 and hADSSL1 were found to be significantly more abundant in DMD patients compared with healthy controls in either one or both assay conditions, which is consistent with what was observed for *mdx* mice (Figure 3A, Table S5). hASAH1, hDMD and hDDX6 were less abundant in DMD patients compared with healthy controls. In addition, the trend for hASAH1 was opposite to what was observed in mice. Six of these proteins (hTPI1, hMYL3, hASAH1, hTTN1, hMSTN and hDDX6) were also differentially abundant in BMD patients compared with DMD (Table S5). Differences in trajectories over time between DMD, BMD and healthy controls were further studied using a linear mixed effects model. hMYL3 had a steeper decline with time form first collected sample in DMD patients compared with BMD and healthy controls (Figure 3B, Table 4).

4 | Conclusions

In this study, we identified plasma biomarkers that are associated with the effect of microdystrophin therapy in *mdx* mice and subsequently evaluated translation to DMD. Out of 83 proteins, 10 proteins were found to be significantly more abundant in plasma from dystrophin-deficient *mdx* mice compared with wt mice and correlated with increasing dystrophin expression upon treatment with the SGT-001 therapeutic. These biomarker candidates, if validated in the future, can potentially be used as exploratory biomarkers for therapies, which aim to restore or replace dystrophin expression. For 5 of these 10 proteins, the trajectory changes were associated with expression of dystrophin in TA muscle of BMD patients. Furthermore, the four proteins hASAH1, hCA3, hDMD and hADSSL1 could also separate

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DMD patients from healthy controls where all but hASAH1 displayed similar abundance differences as when comparing *mdx* mice to wt mice.

The plasma proteome contains proteins spanning 10^{12} orders of magnitude in concentration, from albumin, present at about 35–60 g/L, to inflammatory proteins such as interleukins, which are present at the ng/L or pg/L scale [36]. When using antibodies in antibody array technologies to study a subset of the plasma proteome, it is assumed that the protein being targeted is present at a concentration within the detectable range of the antibody and that a reporter molecule is incorporated into the target protein at a high enough concentration to obtain a sufficient signal-to-noise ratio. In previous studies using antibody SBAs [25], human serum or plasma was biotin labelled at an estimated biotin-to-protein ratio of 1:9 and assayed at a 1:700 sample dilution, which does not account for the difference in detection range for the specific antibodies used. In this study, two optimal assay conditions were then picked from these data to maximize the cover of antibody dynamic ranges. Using these two experimental conditions, a total of 20 biomarker candidates were found to correlate with dystrophin expression in untreated and SGT-001 treated *mdx* mice, with mTTN having the strongest separation between the two groups and strongest correlation to dystrophin expression. PCA revealed that mTTN and mMYL3 contributed the largest to separating *mdx* mice from wt. The present study confirms previous studies reporting elevated levels of serum mMYL3 and plasma mTTN in *mdx* mice compared with wt mice [37–39]. In addition, the observed correlation between dystrophin expression and plasma mTTN levels is partially supported by reports regarding the decrease of N-terminal mTTN in plasma of *mdx* mice and *mdx* dogs at microdystrophin expression exceeding 50% of the muscle fibres in quadriceps and vastus lateralis, respectively [37]. Both mTTN and mMYL3 play an important role in muscle function. While mTTN is involved in the assembly, maintenance and stability of the sarcomere, mMYL3 is responsible for the regulation of muscle contraction. The absence of dystrophin destabilizes the sarcolemma and the transfer of sarcomeric contraction forces to the extracellular matrix [40].

Changes in longitudinal serum trajectory of hMYL3 between DMD and BMD patients hint at a preserved relationship to dystrophin expression or possibly muscle leakage even in humans, although the results are preliminary. Previous studies showed that expression of hTTN and hMYL3 in muscle cells was reduced in the absence of dystrophin whereas expression of hCA3 was unaffected by dystrophin deficiency [41]. Recent studies have identified decreased levels of dystrophin in plasma from DMD patients [35]. A trajectory change in blood could therefore indicate a rescue of tissue integrity in the *mdx* mouse model and in humans. The analysis showed that serum hMYL3 had a detectable trajectory change between DMD and BMD patients in a longitudinal natural history cohort, with steeper decline of hMYL3 in DMD patients over time than what was observed in BMD patients. Although preliminary, this indicates that hMYL3 would be interesting to consider as a gene-therapy pharmacodynamic biomarker. However, no significant trajectory change was observed between BMD patients with low dystrophin expression compared with higher expression for hMYL3. It is, however, important to consider that in the linear mixed effects model, eight of the patients had one sample collected up to 7 years prior to the other samples and the models are therefore assumed to get disproportionally influenced by these older samples. Dystrophin expression was also assumed to be constant during the study period because we have previously shown that dystrophin expression in TA, when measured with Capillary Western immunoassay, does not change significantly in up to 5 years in BMD patients [21]. However, these results are only hypothesis driving, and further studies with more evenly distributed longitudinal samples are therefore needed to confirm these observations.

TTN isoforms range between 625 and 3700 kDa depending on the tissue origin [42] and is part of the sarcomere complex, responsible for muscle contraction. Fragments of titin have gained interest as biomarker candidates for DMD [42] as hTTN has been shown to get proteolytically cleaved in dystrophic tissue, possibly through a mix of tissue-specific calpain and matrix metalloproteinase activity, before leaking into blood. N-terminal proteolytic fragments of hTTN have been detected in both blood and urine, and N-terminal TTN levels have been shown to decline with age in DMD patients [42]. There are also indications of a central, 170–240 kDa large fragment being present in blood from both *mdx* mice and young patients with DMD, that decline rapidly with age [1]. The antibody used in this study, HPA030048, was produced against amino acids 2229–2352 of the human skeletal muscle isoform. To the best of our knowledge, this sequence has not yet been described as a potential blood biomarker. The lack of concordance between human and mouse TTN blood levels in our study could be explained by a difference between proteolytic products of TTN in mouse plasma and human serum. Detected N-terminal hTTN and mTTN fragments in urine differ slightly in size and abundance [43], suggesting a similar difference in blood. We have therefore undertaken further studies to map and characterize the TTN fragments that can be detected in *mdx* mouse and human DMD and BMD patients.

Although our study identifies hMYL3 and hTTN as promising biomarkers, it is important to point out that the *mdx* mouse does not entirely recapitulate the human disease progression besides not being representative for the genetic variation encountered in the patient population. The *mdx* mice exhibit severe muscle degeneration and defects in cardiac function later in life during adulthood [44] compared with the DMD patients that experience the disease onset during childhood. Furthermore, as both biomarker candidates are expressed in heart and skeletal muscle, they might also reflect species specific discrepancies due to the differences between the mouse and human DMD phenotype. Translation of these results from mouse to patients is investigated using samples from untreated DMD patients and BMD patients. Although the microdystrophin-restoring therapy aims to convert the severe DMD phenotype into a mild BMD phenotype, the samples used to confirm these biomarkers are not donated by treated patients but originate from DMD and BMD patients within different age range. Although both hTTN and hMYL3 are promising biomarker candidates, translation from the mouse model to human context will require additional studies in particular using samples donated by gene therapy-treated DMD

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patients. Furthermore, development of robust, sensitive and accurate biomarker quantification assays is paramount for translation of these biomarkers to clinical use.

In conclusion, this study highlights 10 biomarker candidates, which correlates with improved microdystrophin expression in mice and where a decrease of these proteins in blood could indicate a rescue of tissue integrity in the *mdx* mouse model. We further show preliminary indications that hTTN, hMYL3, hDMD, hCA3 and hADSSL1 could be interesting to study further as exploratory endpoints in clinical trials on gene therapy.

Author Contributions

C.J., J.F.B., K.J.B., P.S. and C.A.K.S. conceptualised and designed the study. C.J. performed the biomarker analysis and visualised the results. J.F.B. and K.J.B. performed the mdx study. Z.K., N.v.d.V., J.J.G.M.V., E.H.N. and P.S. designed and performed the clinical study. C.A.K.S., P.S. and E.H.N. secured funding. C.J. and C.A.K.S. wrote the original draft. All authors reviewed the manuscript.

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Conflicts of Interest

Jessica F Boehler and Kristy J Brown are employees and hold equity from Solid Biosciences, Inc. Other authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

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